

Language-independent specification

A **language-independent specification (LIS)**^[1] is a programming language specification providing a common interface usable for defining semantics applicable toward arbitrary language bindings.

LIS's are language-agnostic; they mitigate the risk that a certain language binding might reduce compatibility with other languages. An ideal LIS allows the language bindings to take advantage of features of a programming language uncompromisingly.

Examples of LIS include Interface description language (IDL), Simplified Wrapper and Interface Generator (SWIG) and Common Language Infrastructure (CLI).

Recursive transcompiling can be used to distribute a language independent specification across many different technologies, with each technology potentially keeping an authoritative description of a different part of the specification. Recursive transcompiling provides the general methodology for distributing this authoritative information through the rest of the derivative code pipeline.

See also

- Derivative code
- ISO/IEC 10967
- ISO/IEC 11404
- Language interoperability
- Pivot language
- Recursive transcompiling
- Specification language
- Transcompiler

References

1. "Information technology - Open Systems Interconnection (OSI) abstract data manipulation - Application Program Interface (API) (Language independent)" (<https://cdn.standards.iteh.ai/samples/24130/cf123942dbc7489798a49e85662c5c5f/ISO-IEC-14360-1996.pdf>) (PDF). *cdn.standards.iteh.ai*. Retrieved December 29, 2025.

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